

Scale-dependency of hydraulic conductivity: an example from Thorne Moor, a raised mire in South Yorkshire, UK

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Abstract:

The hydraulic conductivity (K) of many low permeability materials is strongly scale-dependent. In raised mires and other types of peat deposit the effects of features such as abandoned infilled ditches, root holes and localized woody material, cause K to be heterogeneous and scale-dependent. Despite this, field measurements are routinely made using auger hole (slug) tests at the scale of only a few tens of centimetres. Such measurements are locally valid, but where the regional subsurface movement of water through peat bogs is simulated using groundwater models, typically at the scale of hundreds of metres, they give rise to a systematic underestimate of flows and an overprediction of water table elevations. Until now, techniques to obtain values at a scale sufficiently large to include the effects of localized features of higher permeability have not been applied routinely.

Research at Thorne Moor, a large raised mire, demonstrates that the K of peat varies over several orders of magnitude when measured at different scales, using a variety of techniques. Laboratory and auger hole tests cannot be relied upon to provide results that represent the hydraulic conductivity of large expanses of peatland. This has significant implications for the management and long-term restoration of peatlands where both regional and local control of water levels is crucial. For groundwater models to be used successfully to plan such schemes, it is essential to apply the K values relevant to the scale of the simulation.

This paper describes and tests novel techniques, using ditches, for the derivation of K at large scales which overcome many of the problems that have been identified with conventional techniques and are capable of producing estimates that are appropriate to the application of physically based regional flow models. Copyright © 2004 John Wiley & Sons, Ltd.

KEY WORDS peat; raised mire; hydraulic conductivity (K); scale; Thorne Moor

BACKGROUND

Raised mires are characteristically domed peat deposits standing up to 10 m above the surrounding land. The peat accumulates over time, growing upward and outward to sustain a domed water table, which closely follows the shape of the ground surface. This groundwater mound is rain-fed and maintained by the low hydraulic conductivity (K) of the mass of the bog, which restricts the outflow of water (Bromley and Robinson, 1995). Raised bogs are less widespread in the UK than other mire types, but because of their uniform quality and ease of working they have been more subject to human pressures including drainage for agricultural use, cutting for horticulture and for use as fuel. Over 95% of the original 85 000 ha of raised mires in the UK have been damaged or destroyed since 1850 (RSNC, 1990).

The threat to the continued existence of this type of wetland has led to concerted efforts to protect existing remnants, and where possible restore areas that have been subject to large-scale, mechanized peat extraction (Money, 1995; Wheeler and Shaw, 1995). This has now been considerably strengthened under legislation resulting from the European Community's *Habitats Directive*. Properly designed protection and restoration

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schemes, however, depend critically upon an understanding of groundwater flow, which is largely dependent upon the K of the peat.

A problem encountered with investigations of such peatlands is that when direct field measurements are used to obtain model parameters they are often unable to produce satisfactory results. Kneale (1987), working on the Mongan Bog in central Southern Ireland, found that the predicted water-table mound for this relatively undisturbed bog was overestimated and that its shape was best fitted using a value of K an order of magnitude larger than that obtained by site measurements using the auger hole technique. She argued that this 'effective' areal K was greater than those obtained from point measurements owing to water movement through the open pool network. Similarly, Bragg (1991) working at Wedholme Flow in Cumbria, found that it was not possible to simulate the observed water levels with K values measured on the bog using the auger hole method. Instead she was forced to arbitrarily increase the conductivity values by two orders of magnitude to 'fit' the observed water levels.

To obtain acceptable model predictions the authors had to back-calculate 'effective' values of K that were much higher than their field measurements. Although having no direct measurements to support or test their claims, both authors attributed the problem to the auger hole method providing only a measure of the matrix permeability of the peat, and not allowing for preferential flow paths through pools or peat voids. In these areas the apparent scale-dependency of K was assigned to the impact of natural heterogeneity; in other areas, however, the effects of human activity, such as cutting drainage channels, may also be a contributory cause. Such apparent scale-dependency is not restricted to peat sites, and has a much wider significance; Beven (1989), for example, has noted a similar problem in descriptions of river catchment modelling studies at a number of non-peat sites.

The inability to reconcile values obtained by means of the most commonly used field technique with the requirements of regional flow models prompted the current investigation into the extent to which K is dependent upon the scale of measurement.

This paper describes the results of field studies in a raised mire to measure the K of peat at a range of scales.

STUDY SITE

The work was undertaken at Thorne Moor in South Yorkshire, chosen because it is a wetland with some existing hydrological data, has a history of peat cutting and is an area for which restoration plans are currently under way. Thorne Moor forms part of the Humberhead levels and at 19 km² is the largest single remaining area of intact raised mire in England. It lies in one of the lowest rainfall parts of the country and is thus particularly at risk from any disturbance likely to dewater the peat. The moor has been subject to agricultural encroachment and to hand cutting for peat over several centuries, particularly from the 1870s onwards (Smart *et al.*, 1986). In the past 20–30 years, however, large-scale mechanized commercial peat harvesting has taken the place of hand cutting. Since the late 1980s much of the northern and eastern parts of the mire were mechanically milled, leaving the southern, central and western sections relatively untouched. The entire moor has long been a Site of Special Scientific Interest (SSSI), but recently the 9.75 km² of mire that had not been mechanically milled became a candidate Special Area of Conservation (cSAC), an area which includes the 0.7 km² Thorne National Nature Reserve (NNR) established in 1987 (Figure 1).

Although the cSAC has not been mechanically milled it has been subject to hand cutting from the late nineteenth to the middle of the twentieth century (Caufield, 1991). The pattern of the abandoned rectangular system of drains, trenches and intervening uncut baulks constructed during traditional hand cutting remains imposed on the topography, a feature that is clearly visible on aerial photographs. This cutting partially removed the upper layers, but 1–3 m of peat remain overlying a 4–6 m thick bed of Quaternary clay (Smart *et al.*, 1986). The remaining peat is vegetated with species such as bog mosses (*Sphagnum* spp.), cotton grass (*Eriophorum angustifolium* and *E. vaginatum*), heather (*Calluna vulgaris*) and heath (*Erica tetralix*). Such limited cutting may actually enhance the conservation value of the mire by increasing the range of

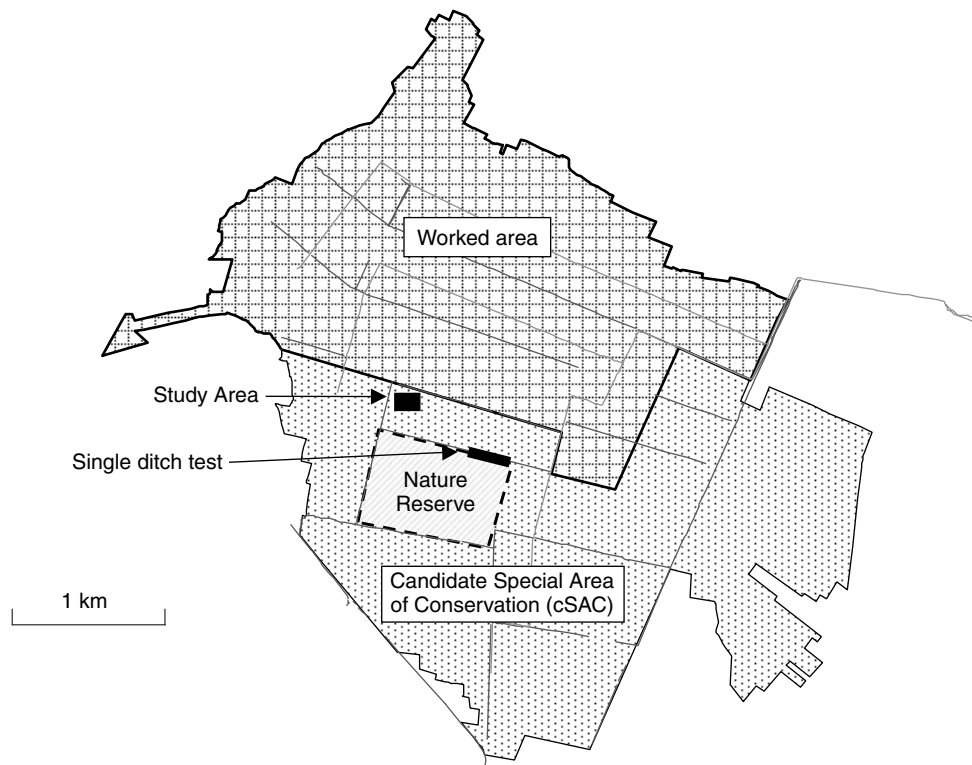


Figure 1. Thorne Moor, South Yorkshire, showing the cSAC area, the mechanically worked area, the National Nature Reserve and the study area. Major internal drains are indicated, but not the myriad of smaller drainage pathways

habitats, and it must be differentiated from the much more intensive and environmentally destructive modern commercial surface milling practices where all of the surface vegetation is completely removed to expose the peat, and a thin surface layer is then skimmed off every few weeks through the summer.

English Nature, the UK government's agency for wildlife conservation, is currently working to initiate regeneration of areas that until recently have been worked. As part of the conservation effort a series of groundwater modelling studies were undertaken at Thorne to assess the impact of peat working on water levels in the conservation area (Bromley and Robinson, 1991; Robinson *et al.*, 1992; Bromley *et al.*, 1997). Hydraulic conductivity was initially represented in the models using site-measured values obtained from auger-hole techniques (Newson, 1987; Heathwaite, 1990). However, it proved impossible to simulate the observed water tables using these values; acceptable simulations required permeability values that were one to two orders of magnitude higher (Bromley *et al.*, 1997).

CONTROLS ON PEAT HYDRAULIC CONDUCTIVITY

In an undisturbed natural raised mire the peat profile can be categorized broadly into two layers, an upper 10–40 cm thick aerated layer of relatively undecomposed plant material, known as the *acrotelm*; and below this an anaerobic zone of waterlogged and more humified peat termed the *catotelm*, which normally makes up the bulk of the peat deposit (Ingram, 1982; Hobbs, 1986; Bragg, 1991). The K in the acrotelm is typically several orders of magnitude higher than the catotelm, where values are uniformly low. In undisturbed mire the catotelm is permanently saturated whereas within the acrotelm, which provides a thin but high permeability pathway for water movement, the water table fluctuates seasonally. In the case of disturbed moors, such as

Thorne, cutting of peat has removed much of this high permeability surface layer so that the local and regional K of the site now tends to be dominated by the catotelm.

Although the removal of the acrotelm has reduced the *average* permeability of the peat profile the cutting had the effect of increasing the *regional* permeability by creating a network of drains, which criss-crossed the region in a rectilinear pattern. Following the end of hand cutting in the early part of the twentieth century many of the smaller ditches either collapsed or were partly infilled by vegetation and peat growth, creating localized subsurface pathways for preferential water movement. The presence of such pathways has tended to increase the K of the peat at the scale of the main ditch network.

The presence of undecomposed fibrous or woody material within a peat profile can also play an important role in controlling local K . At Thorne Moor this takes the form of scattered concentrations of woody or fibrous material, associated with the remains of individual trees or large shrubs, generally birch, oak or pine (P. Roworth, English Nature, personal communication, 2002). The presence of a concentration of this type of material provides pathways for preferential water movement and can significantly increase the local K (Maggs, 1997).

METHODS

Four measurement techniques, each sampling a different volume of peat, were used to test for a link between K and scale.

1. Laboratory measurements on cores: sample volume *c.* 0.002 m³.
2. Auger hole tests: sample volume *c.* 0.4 m³.
3. Double-ditch (*c.* 10 m in length), steady state tests: sample volume *c.* 30 m³.
4. Single-ditch (*c.* 400 m in length), transient tests: sample volume *c.* 360 m³.

Laboratory measurements and auger hole tests are the most widely used methods, but they sample only a limited volume of material. In contrast, field tests based on the use of ditches that sample much larger volumes, but are considerably more difficult to apply in practice, have not been used routinely.

In this study the first three methods were applied at a field site located to the north of the NNR, while the much longer single ditch test and additional auger hole tests were conducted on the northern boundary of the NNR (Figure 1).

Laboratory measurements

Three cores, 90 cm long, were obtained by inserting a sharpened metal pipe (15.2 cm internal diameter) vertically into the peat surface and digging down beside the pipe to retrieve the core. Each core was then cut into 10-cm lengths, providing a total of 27 samples. In the laboratory, saturated K measurements were made with a constant-head permeameter (e.g. Wang *et al.*, 1985).

Auger-hole or 'slug' tests

Slug tests are a basic investigative tool for studying the permeability of materials (van der Kamp, 2001). In this study small-diameter holes were carefully augered by hand to the base of the peat, and then lined with 25.4 mm internal diameter slotted PVC pipe to within 30 cm of the ground surface. The top 30 cm was lined with blank casing and sealed with bentonite. A 'slug' of water is either instantaneously added to or removed from the hole and the subsequent rate of water level decline, or recovery, is recorded. This information, together with details of hole-geometry is used to calculate K . At the study site (Figure 2), auger hole tests were conducted on 35 holes spaced mostly 3 m apart and arranged in a grid pattern. A series of three separate tests were made at different times of the year (March, July and September), to capture the full seasonal range

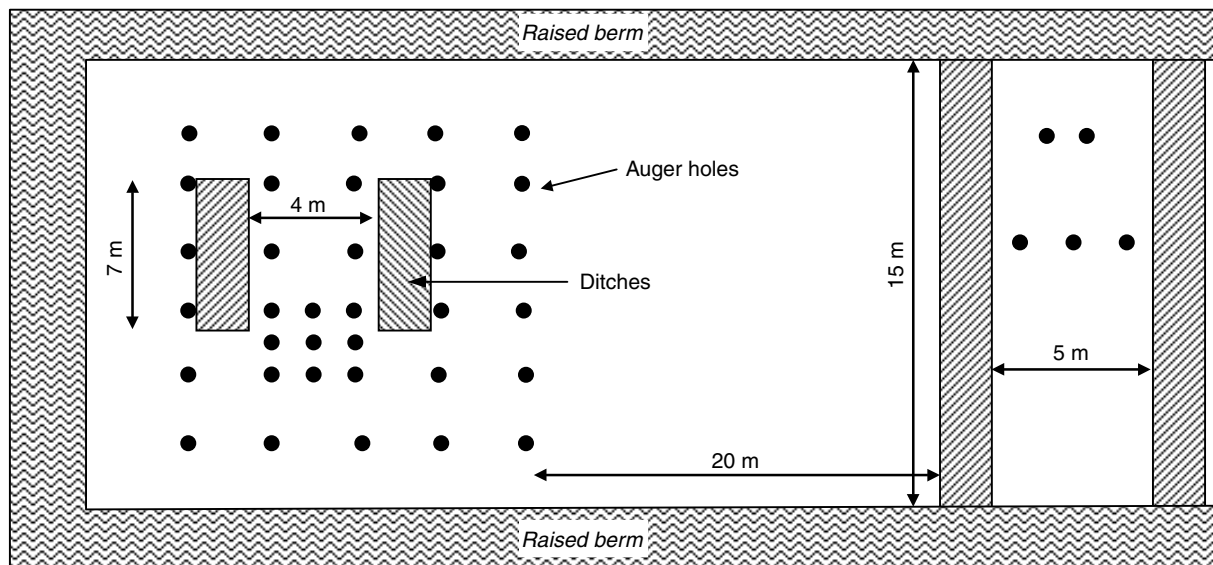


Figure 2. Study area for the auger-hole and double-ditch tests. The laboratory samples were taken from the area between the ditches

of water-table conditions. Falling head tests were used to calculate K following procedures recommended by the British Standards Institute (BSI, 1981). The fall of head for each test was monitored at 60-s intervals using a multichannel Campbell CR10 logger. Based on the work of van Beers (1983) the effective diameter of the measured column was taken to be 80 cm. Additional independent auger test measurements of K at this site are published (Newson, 1987; Heathwaite, 1994).

Double-ditch tests

This technique was applied at the site of the auger hole network (Figure 2), and required the excavation of two parallel ditches to the base of the peat to expose the underlying impermeable clay. Following excavation, ditch water levels were then allowed to reach equilibrium before the test began. For these tests water was continuously pumped from one ditch to the other to establish a constant head difference between the two (Figure 3). The pumping rate was adjusted until a steady state condition was established with a constant head difference at a constant flow. At this point the subsurface flow through the intervening peat between the ditches has a constant relationship to the rate at which water is being pumped from one ditch to the other. Groundwater levels between the ditches were monitored in piezometers to confirm that a steady state condition was achieved. Hydraulic conductivity was calculated using an equation developed by Barker (1986) to describe flow between two parallel fractures in hard rock. The ditches can be considered analogous to this type of system, both having the same geometry. The transmissivity (i.e. permeability multiplied by aquifer thickness) can then be calculated from the following equation

$$T = \frac{Q}{2\pi s_0(0.25 + 0.83b/a)} \quad (1)$$

where T is transmissivity ($\text{m}^2 \text{ day}^{-1}$), Q is discharge ($\text{m}^3 \text{ day}^{-1}$), b is the length of ditches (m), a is the distance between ditches (m) and s_0 is the drawdown in the donor ditch (m).

Two sets of pumping tests were carried out using different lengths of ditches. In the first case the ditches were 7 m long, 2 m wide and 4 m apart; in the second they were 15 m in length, 2 m wide and 5 m apart. Pressure transducers connected to a data logger recorded water levels in both ditches; water levels in the

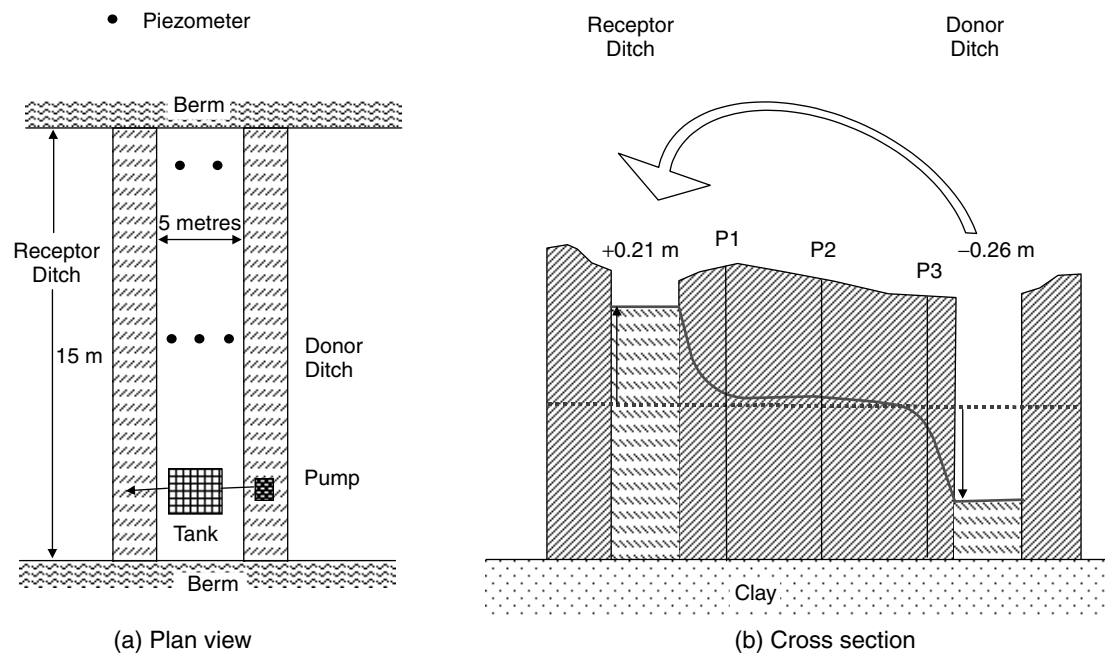


Figure 3. (a) Plan of 15 m double-ditch test; (b) cross-section showing steady state conditions at the end of the test

piezometers were measured manually. A control piezometer, located 20 m from the ditches, was monitored with a transducer and logger to provide a record of natural groundwater level changes during the test period. During the tests, water was pumped from the donor to the receptor ditch using a dirty-water submersible pump via a header tank. Flow into the receptor ditch was kept constant using a gate valve.

Single-ditch analyses

There are practical limits to the size of ditches that can be used for the double-ditch system, owing to the volumes of water that have to be continuously pumped at an isolated site without a power source and the increasing lengths of time involved.

At the largest scale of measurement (several hundred metres in length) an alternative technique was developed, using data from an existing ditch located 800 m to the south-east (Figure 1). The data used were obtained from an earlier study undertaken in 1990 (Heathwaite, 1994). This was part of an experiment to test the feasibility of a ditch recharge system protecting the NNR from the dewatering effects of adjacent peat cutting, by maintaining high water levels in a boundary ditch. High water levels were maintained in that ditch by pumping water from a nearby 400 m long 'donor' ditch. Pumping was not continuous, but took place intermittently in a series of 26 discrete pumping events each lasting on average 1.75 h.

For each pumping event the water level in the donor ditch was lowered by up to 30 cm before the pump was switched off, and the water level was allowed to recover. The effect was analogous to an auger hole 'bailing' test in which water is instantaneously removed from a hole and allowed to recover, but conducted at a much larger scale and with a different geometry. To obtain the bulk K of the peat from these data a three-dimensional finite element groundwater model was used to simulate the recovery for a number of selected pumping events. The K was optimized so as to minimize the difference between the observed and simulated water level response. Although not as rigorous as the steady state double-ditch approach, the single-ditch method provides a reasonably reliable estimate of K and is feasible for much larger lengths, and hence is appropriate to much larger scales, than would be practical using the double-ditch technique.

There are also conductivity data obtained from auger hole tests carried out adjacent to the donor ditch from other studies by Newson (1987) and Heathwaite (1994), which augment those undertaken during the present study.

RESULTS

The results of the measurements are described in order of increasing scales.

Laboratory measurements

The smallest scale measurements, made in the laboratory (sample volume *c.* 0.002 m³), are summarised in Figure 4. This profile represents an average of three cores sampled at 10-cm intervals from the peat surface (Lee-Preston, 1999). Hydraulic conductivity throughout the upper 90 cm is relatively uniform, falling between 0.1 and 0.4 $\times 10^{-5}$ m s⁻¹, except for the uppermost 10 cm where the figure increases by an order of magnitude to over 4.2 $\times 10^{-5}$ m s⁻¹. However, these higher values close to the surface are to be expected because at this level *Sphagnum* decomposition and compaction is at a minimum. This upper zone is thin because decades of peat cutting have removed much of the higher more permeable parts of the natural bog, leaving behind the lower less permeable horizons. The geometric mean for the full 27 laboratory measurements is 2.4 $\times 10^{-6}$ m s⁻¹, reducing to 1.95 $\times 10^{-6}$ m s⁻¹ if the three samples from the upper 10 cm are excluded (Figure 5a).

Auger-hole or 'slug' tests

At a larger scale (sample volume *c.* 0.4 m³) the results of 93 auger-hole tests performed on 35 auger holes at three different times of the year are presented in Figure 5b. As in the case of laboratory measurements there is a wide range of values; the individual auger hole measurements ranged from 1.31 $\times 10^{-7}$ up to 8 $\times 10^{-6}$ m s⁻¹. The geometric mean is 1.3 $\times 10^{-6}$ m s⁻¹, which is slightly lower than the laboratory results, but not statistically significantly different. It should be noted that the upper few centimetres of highly permeable peat profile cannot be measured using the auger-hole technique and are thus excluded from the mean figures.

Although there were large differences between tubes, measurements taken at the same tube at different times of the year were relatively consistent so that the overall spatial pattern of variability remained the same.

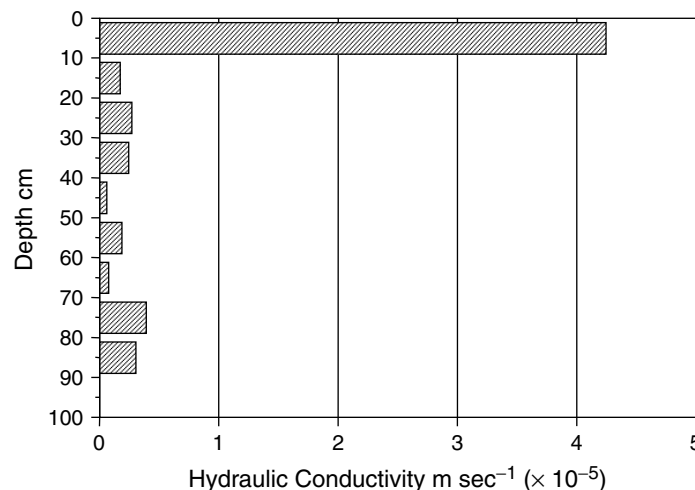


Figure 4. Laboratory measurements of hydraulic conductivity from three cores, showing the mean and range of values over 10-cm lengths from the peat surface

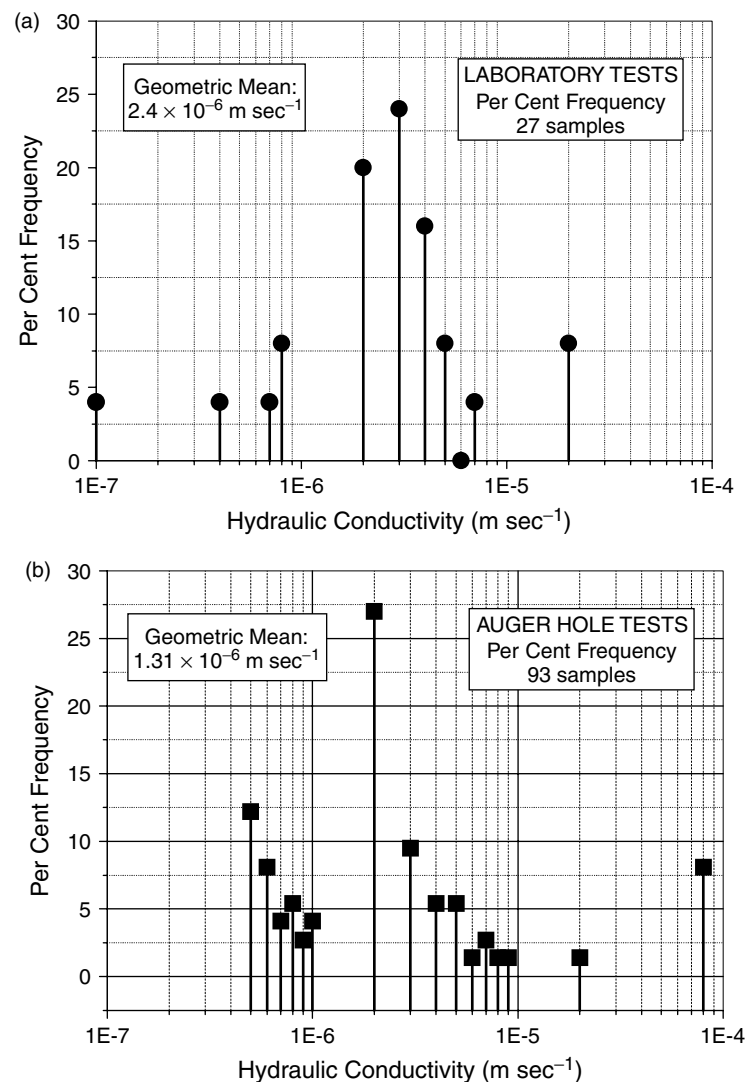


Figure 5. Comparison of hydraulic conductivity values from (a) laboratory measurements and (b) auger-hole measurements

The correlations between the three sets of readings varied between 0.66 and 0.94, with the measurements at individual tubes varying by less than 30% between the different dates. For seven of the measurements the response was too rapid for the 60-s logging interval to obtain a reliable result because the hole directly intercepted a zone of high permeability.

The wide range of values obtained reflects the heterogeneous nature of the peat at the scale of the sample grid. Where features such as increased development of root holes, woody material or infilled ditches are encountered K is high, but these are localized and at the scale of the auger-hole test are encountered only by chance. Such 'aberrant' results are routinely discarded in many studies, although they may be quite valid, and this can introduce a systematic bias in field measurements. In order to calculate an overall geometric mean conductivity value from the auger-hole measurements the K value in these cases was arbitrarily assumed to be two orders of magnitude higher than the geometric mean for the successfully measured tubes. The geometric means of the readings were $9.47 \times 10^{-7} \text{ m s}^{-1}$ in March, $9.85 \times 10^{-7} \text{ m s}^{-1}$ in July and $9.32 \times 10^{-7} \text{ m s}^{-1}$

in September, and are relatively insensitive to the extreme values assumed. The mean values obtained are comparable with the values independently derived by Newson (1987) and Heathwaite (1994).

The results of both the laboratory and the auger-hole measurements are characterized by extreme variability, ranging over three orders of magnitude (Figure 5a and b), and there is no statistically significant difference between them (Wilcoxon two-sample rank test). Clearly, at this scale, the volume sampled by individual tests (less than 0.4 m^3) is well below the 'representative elementary volume' (REV) of the peat body. This is the minimum volume needed to include all those features controlling the variability in K and contributing to the hydraulic conductivity of large expanses of peatlands.

Double-ditch tests

Compared with the auger-hole technique, the double-ditch tests increased the sample volume by about two orders of magnitude. In the first test, the parallel ditches were 7 m long and pumping lowered the water level by 0.3 m in the donor ditch and raised it by 0.21 m in the receptor ditch. The rate of water flow needed to maintain this head difference was less than 0.5 L min^{-1} , which could not be accurately maintained or measured with the pumping equipment available. The length of time required to achieve a steady state circulation between the ditches at this flow rate would have been excessively long and in any event would probably have been masked by natural water level fluctuations resulting from rainfall or evaporation. All indications suggested a low K with no major pathways of higher permeability between the ditches. As a result the pumping was abandoned after 3 days and instead a time-varying solution was obtained using the same technique of ditch water recovery applied to the single-ditch tests. The subsequent result, $6 \times 10^{-6} \text{ m s}^{-1}$, is somewhat higher than the mean obtained for auger-hole and laboratory tests.

In the second test, using ditches 15 m in length, water was pumped at a constant rate of $2.4 \times 10^{-4} \text{ m}^3 \text{ s}^{-1}$. Pumping continued for a period of 2.5 days until steady state conditions were established (Figure 3b). This induced a drawdown of 0.26 m in the donor ditch and a total head difference of 0.45 m. Once the variables, Q and s_0 , were determined, solution of Equation (1) for the known ditch geometry gave a transmissivity of $5.4 \times 10^{-5} \text{ m}^2 \text{ s}^{-1}$. Taking the average saturated thickness of peat between the two ditches to be 0.5 m, the transmissivity converts to a K of $1.1 \times 10^{-4} \text{ m s}^{-1}$, two orders of magnitude higher than results obtained from auger-hole and laboratory measurements.

During the test it was noted that a significant part of the flow into the donor ditch was provided by inflow from a small root hole situated just above the pumping water level. The inflow was measured volumetrically and found to be a constant $7.5 \times 10^{-5} \text{ m}^3 \text{ s}^{-1}$. This is approximately one-third of the total flow between the ditches and confirms the potential importance of local features such as root holes on regional permeability. Without this pipeflow component the calculated K reduces from $1.1 \times 10^{-4} \text{ m s}^{-1}$ to $6.9 \times 10^{-5} \text{ m s}^{-1}$.

Single-ditch analyses

For this largest scale test, using the 400 m long ditch, the recovery of water levels following four selected pumping events were simulated using a three-dimensional finite element groundwater model. The pumping events took place in July, August, November and December 1990 and each lasted between 1 and 3 h. The two simulations shown in Figure 6 illustrate the full range of solutions; the first is for an event in August (6a), the second for one in November (6b). In each case the observed water level recovery is matched against a series of modelled recovery curves for a range of K values. In the first example, for the August event, the best match produces a value of $8 \times 10^{-4} \text{ m s}^{-1}$; for the November event the K value was $1 \times 10^{-4} \text{ m s}^{-1}$. Within this range the four solutions produced a mean of $4 \times 10^{-4} \text{ m s}^{-1}$. In a transient solution, the specific yield of the peat is required. As this was not known a sensitivity analysis was conducted, using a range of values, to assess the impact of this factor on the result. Figure 6 demonstrates that the envelope of the curves produced (shown here for two extreme values, 10% and 30%) is relatively narrow and does not prevent a reliable match. It is interesting to compare these results with the hydraulic conductivity estimates of previous work at this part of the moor, which had indicated values of about $2 \times 10^{-7} \text{ m s}^{-1}$ (Newson, 1987) for 100–150 cm deep

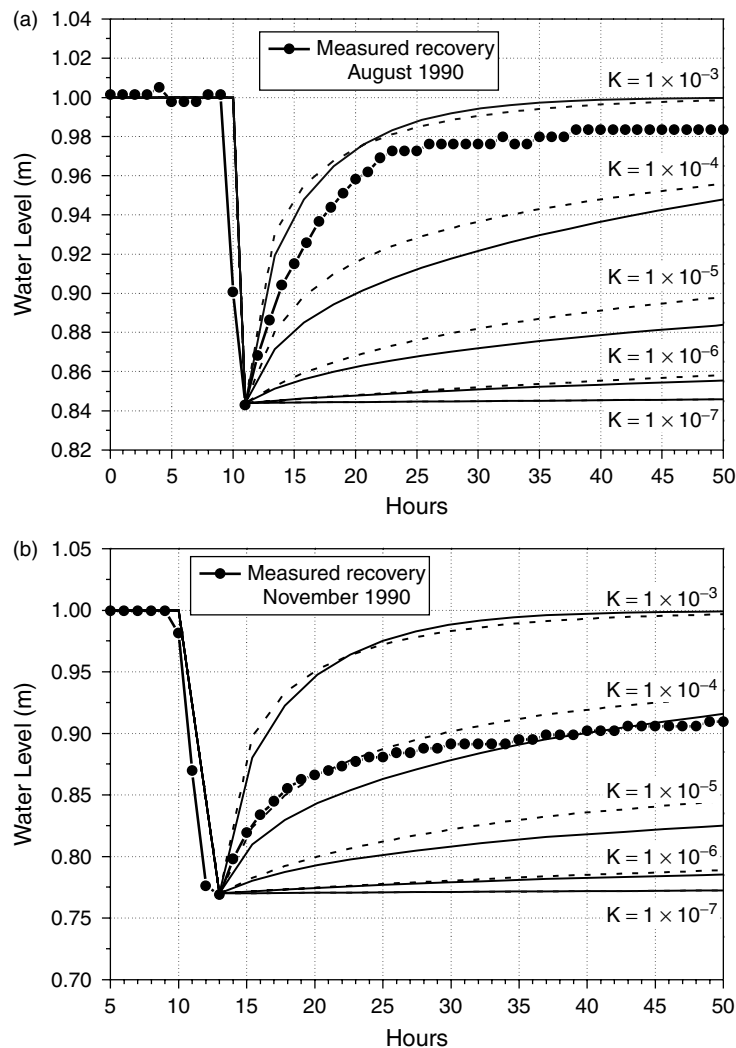


Figure 6. Observed and simulated ditch water level recovery after pumping from a 400 m single ditch on (a) August 1990 and (b) 27 November 1990. Each modelled K curve is given for two extreme values of specific yield; 10% (solid line) and 30% (dashed line)

piezometers and about $2 \times 10^{-5} \text{ m s}^{-1}$ to $5 \times 10^{-6} \text{ m s}^{-1}$ for slightly shallower depths (Heathwaite, 1994). These auger-hole test results are much lower than the estimates from the single ditch analysis, but are quite comparable with the auger-hole results obtained in the present study at the double ditch site, confirming the similarity of the peat at the two locations.

The results of the whole series of tests is summarized in Figure 7. Measured K increases by almost three orders of magnitude as the scale of measurement is increased from less than 1, to several hundred metres. The highest measured value is comparable to those required when a regional groundwater model was applied to the southern unmilled part of the Moor (Bromley *et al.*, 1997). The results clearly indicate that the regional values required by groundwater flow models should be obtained using field measurements at the scale of hundreds rather than tens of metres. Conventional measurements of hydraulic conductivity from small-scale field tests (auger-hole method) or from laboratory cores are quite inappropriate for the parameterization of regional flow models.

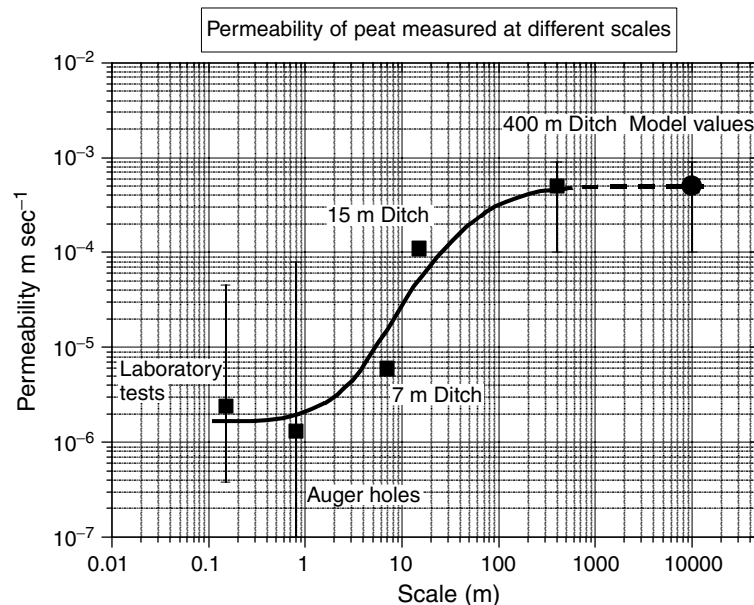


Figure 7. Comparison of peat hydraulic conductivity values, measured at a range of scales, together with the value derived by fitting a regional groundwater model to the observed water table elevations in a network of bore holes across the Moor

DISCUSSION AND CONCLUSIONS

The comparison of hydraulic conductivity measurements at increasing scales has shown a clear tendency for higher values to be obtained when larger volumes of peat are considered. This result is intuitively sensible, as the peat contains roots and woody fragments and is crossed by many ditches (both visible and relict) providing preferential flow routes.

Laboratory and auger-hole tests both provide measures of the permeability of the peat matrix, and given the amount of variability in the results for each technique there is no evidence of a statistically significant difference.

The failure of these methods to provide results that represent the hydraulic conductivity of large expanses of peatlands is an important result because many applications of 'physically-based' models involve the derivation of model parameters from field measurements. It has been widely found that hydraulic conductivity values derived in the field are too low to account for observed flow processes.

As larger scales of measurement are considered in this work, the values obtained converge to the regional value of K obtained by calibration of a groundwater flow model where extensive records of regional water levels are available. Where such data are not available it is important to be able to measure model parameters such as K in the field in a way that is meaningful for the application of a flow model. The findings of this study, that the widely applied auger-hole technique provides values that are several orders of magnitude too low, is therefore of great significance for the application of models for site management.

There are possible alternative explanations for this scale dependency. Most notably, the general validity of the auger-hole technique itself has been questioned in cases where the 'well' has not been properly 'developed' (Butler *et al.*, 1996; Butler and Healey, 1998). In such situations the method would systematically underestimate the true K ; any scale effect noted in comparison with pumping tests could be in part an artefact. Although this may occur in some situations it is felt that in this study the dominant component of the difference between the methods is scale dependency. Care was taken with the installation of the wells and the overlap of the calculated values with the results of the shorter of the ditch pump tests is indicative of good development of the auger holes. There was good agreement between the auger test results and earlier independent studies,

and there is the direct visual evidence of lines of preferential flow. At areas such as Thorne, where the ground is very heterogeneous and there are numerous flow channels, a scale-dependency of K is to be expected. If the auger hole method can be prone to problems owing to incomplete well development that is an additional reason to advocate the use of ditch pumping tests.

Mire areas that have been damaged by human activities, such as parts of Thorne Moor, now face a long restoration period, which will call for good, well-founded management strategies. To provide effective management requires a sound understanding of the hydrology and in particular of the peat K , which to a large extent controls water movement within the mire. To obtain realistic regional values for K the current work has demonstrated that results from standard techniques based on auger-hole or laboratory tests need to be treated with caution, because they are not appropriate to heterogeneous materials (Figure 7).

Groundwater models provide one of the more effective management tools for peatlands, and it is important to recognize that if inappropriate K data are used, this can produce results that are very misleading and may lead to inappropriate actions. Use of auger-hole or laboratory-derived K in models may lead to underestimates of the actual hydraulic conductivity of the peat and hence will underpredict the extent of any water-table drawdown resulting from influences such as agricultural drainage or peat cutting in nearby areas. This could result in a serious underestimation of the amount of effort and resources required to achieve peatland restoration and an overconfidence of conservation potential of an area. When inappropriate K values are used, physically based groundwater models become little better than simple empirical representations and just as liable to produce misleading results. Successful management based on model predictions thus can be expected only if realistic estimates of regional K are used as input.

This study has shown that the ditch pumping techniques described here provide K values that are far more relevant to the application of flow models than results of conventional auger-hole tests. This conclusion applies irrespective of whether the differences result from the scale dependency of K , as at this site, or to problems sometimes encountered with the application of the auger-hole method. The double-ditch method proposed here is the more accurate of the two pumping methods, but requires continual measurements which can extend over a number of days in low conductivity material before a steady state equilibrium is achieved. In situations where sufficient resources are not available for its application then the single-ditch method may provide a more practical alternative, because it can be applied to an existing isolated ditch and it is much quicker to obtain a measurement.

ACKNOWLEDGEMENTS

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